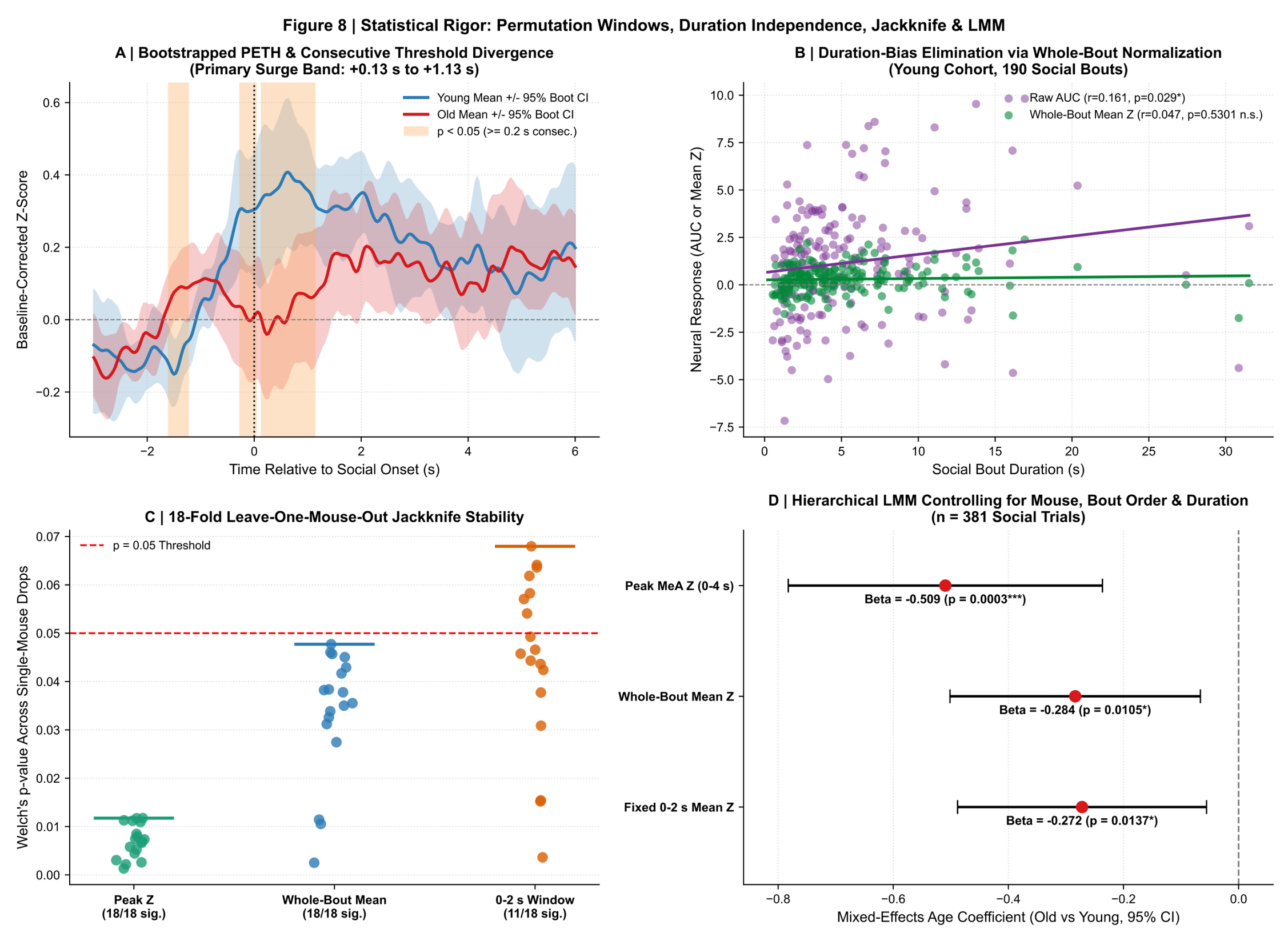

Figure 8 | Statistical Rigor: Permutation Windows, Duration Independence, Jackknife & LMM
A | Bootstrapped PETH & Consecutive Threshold Divergence
(Primary Surge Band: +0.13 s to +1.13 s)
Young Mean +/- 95% Boot CI
Old Mean +/- 95% Boot CI
p < 0.05 (>= 0.2 s consec.)
Baseline-Corrected Z-Score
Time Relative to Social Onset (s)
B | Duration-Bias Elimination via Whole-Bout Normalization
(Young Cohort, 190 Social Bouts)
Raw AUC (r=0.161, p=0.029*)
Whole-Bout Mean Z (r=0.047, p=0.5301 n.s.)
Neural Response (AUC or Mean Z)
Social Bout Duration (s)
C | 18-Fold Leave-One-Mouse-Out Jackknife Stability
p = 0.05 Threshold
Welch's p-value Across Single-Mouse Drops
Peak Z
(18/18 sig.)
Whole-Bout Mean
(18/18 sig.)
0-2 s Window
(11/18 sig.)
D | Hierarchical LMM Controlling for Mouse, Bout Order & Duration
(n = 381 Social Trials)
Peak MeA Z (0-4 s)
Beta = -0.509 (p = 0.0003***)
Whole-Bout Mean Z
Beta = -0.284 (p = 0.0105*)
Fixed 0-2 s Mean Z
Beta = -0.272 (p = 0.0137*)
Mixed-Effects Age Coefficient (Old vs Young, 95% CI)